

INTERNAL ENGINEERING DOCUMENT

FSTS

FSTS Website Operating System(TM)

Improvement Roadmap & Execution Plan

83

Total Improvements

15

Categories

5

Execution Waves

18

P0 Critical

This document converts all 83 approved improvements into a prioritized, wave-based engineering execution plan for the FSTS-WOS(TM) platform. Items are organized by functional area, assigned a priority tier (P0-P3), and placed into execution waves aligned to the Corsair launch milestone and Website #2 readiness goal.

Architecture Principle: Every improvement must be evaluated for reuse across many independent client websites. The FSTS-WOS(TM) is not built around Corsair -- it is a multi-tenant platform that must support service businesses, e-commerce stores, training companies, membership organizations, and any future business type without changing core dashboard code.

Generated August 18, 2026

FSTS Client Dashboard -- For FSTS Team Members Only

EXECUTIVE SUMMARY

Priority Distribution & Execution Waves

P0 -- CRITICAL

18

22% of total

P1 -- REQUIRED

41

49% of total

P2 -- IMPORTANT

18

22% of total

P3 -- ENHANCEMENT

6

7% of total

Wave 1 -- Corsair Critical

Security, payments, authentication, email, live CMS, class/event correctness

? Milestone: Corsair First-Site Production Completion

36 items ? 18 P0

Wave 2 -- Website #2 Readiness

Multi-tenant safeguards, onboarding reliability, reusable seeding, permissions, CI/E2E

? Milestone: Website #2 Ready

32 items

Wave 3 -- Client Self-Service

Reduce FSTS manual support requirements

8 items

Wave 4 -- UX & Performance

Loading states, navigation speed, accessibility, dashboard polish

7 items

Wave 5 -- Growth & Enhancement

Marketing, commerce, advanced automation, and non-blocking improvements

0 items

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WAVE 1

Corsair Critical

Security, payments, authentication, email, live CMS, class/event correctness

? Milestone: Corsair First-Site Production Completion

ID	PRIORITY	CATEGORY	IMPROVEMENT	TAGS
#14	P0	Media & Images	Let clients actually use their uploaded images across their whole site	Client UX Revenue
#21	P0	Email & Notifications	Make failed welcome email deliveries visible so admins know when a new registrant didn't get their email	Client UX Reliability
#65	P0	Email & Notifications	Let admins configure their Resend API key from the Email Config page -- not the Convex dashboard	Client UX Automation
#66	P0	Email & Notifications	Confirm a real email reaches the right inbox once Resend is configured for the first client	Reliability Client UX
#85	P0	Permissions & Route Security	Make the guard check smart enough to catch null checks that don't show an access-denied message	Security
#86	P0	Permissions & Route Security	Extend the access-denied guard requirement to any new route category added outside the sites folder	Security Infrastructure
#92	P0	Services	Confirm services created in the dashboard actually appear on the public website	Client UX Reliability
#135	P0	CMS & Live Data	Confirm a price change in the dashboard actually updates the Corsair website within 60 seconds	Client UX Reliability
#159	P0	Permissions & Route Security	Confirm write-path guards work for CRM, forms, and media modules too	Security
#164	P0	Permissions & Route Security	Confirm a client-role user is automatically blocked from every design-locked route	Security
#171	P0	Permissions & Route Security	Confirm the Footer, Email, CRM, and Payments pages can't be unlocked by a client user	Security
#172	P0	Permissions & Route Security	Catch a client user bypassing the navigation update and delete guards too	Security
#193	P0	Courses & Classes	Make sure a client with a stale session can't register for a class that's already been cancelled	Revenue Reliability
#202	P0	Email & Notifications	Alert admins the moment a paid order confirmation email permanently fails to deliver	Revenue Reliability
#203	P0	Payments & Checkout	Confirm Square payments still process correctly end-to-end after the idempotency changes	Revenue Reliability
#204	P0	Payments & Checkout	Harden the provider-agnostic payment webhook with the same idempotency and email-state guards	Revenue Security Reliability
#206	P0	Payments & Checkout	Block checkout when a course or event price is missing or corrupt so no broken order reaches Square	Revenue Reliability

ID	PRIORITY	CATEGORY	IMPROVEMENT	TAGS
#211	P0	Infrastructure & DevOps	Confirm clients can actually sign in once the Cloudflare DNS fix is applied	Client UX Infrastructure
#15	P1	Media & Images	Prevent broken images when a media asset is deleted that's still in use	Client UX Reliability
#20	P1	Infrastructure & DevOps	Make sure Vercel deploys cleanly against the rewritten commit history	Infrastructure
#22	P1	Email & Notifications	Warn admins when the lead alert recipient address looks wrong or bounces	Reliability Client UX
#23	P1	Email & Notifications	Let admins send a test lead alert to confirm their notification email is working	Client UX Automation
#26	P1	Email & Notifications	Fix broken tests in the email suite that are silently hiding regressions	Technical debt Reliability
#27	P1	Email & Notifications	Confirm the notification email override reaches the right address end-to-end	Reliability
#61	P1	Infrastructure & DevOps	Confirm www.fstscientssystem.com actually redirects once the Vercel alias is added	Infrastructure Client UX
#81	P1	Media & Images	Auto-generate alt text when images are uploaded so clients never have empty alt text	Client UX Automation
#82	P1	Media & Images	Confirm image picker flows work end-to-end in a real browser across every module	Reliability
#136	P1	CMS & Live Data	Extend live CMS data to the course detail pages so per-course pricing is also accurate	Client UX Reliability
#137	P1	Services	Seed services data for Corsair so the live services section appears on the website	Client UX
#141	P1	Email & Notifications	Confirm the welcome email fires when a new user is created -- and only then	Reliability
#143	P1	Email & Notifications	Make sure welcome emails link to the correct dashboard URL after deployment	Reliability
#157	P1	Courses & Classes	Confirm courses stay hidden when the agency switches the module off platform-wide	Security Reliability
#181	P1	User Management & Access Control	Catch a stale legacy role grant the moment a permission is renamed or removed	Security
#189	P1	Courses & Classes	Confirm lifecycle alerts in the edit dialog always reflect live registration data	Client UX Reliability
#194	P1	Courses & Classes	Catch a flyer that stays published after its linked class is deleted outright	Client UX Reliability
#195	P1	Courses & Classes	Confirm the lifecycle clock is registered so tick can never be silently skipped	Reliability Automation

WAVE 2

Website #2 Readiness

Multi-tenant safeguards, onboarding reliability, reusable seeding, permissions, CI/E2E

? Milestone: Website #2 Ready

ID	PRIORITY	CATEGORY	IMPROVEMENT	TAGS
#11	P1	User Management & Access Control	Let admins manually unlock a locked portal user account	Client UX Security
#12	P1	User Management & Access Control	Stop other user management pages from leaking the full site list	Security Reliability
#62	P1	Infrastructure & DevOps	Turn on the branch protection rule so the identity check can't be bypassed on GitHub	Infrastructure Security
#72	P1	Onboarding & Site Setup	Add an onboarding wizard so admins can seed a new client without touching the CLI	Automation Client UX
#73	P1	Onboarding & Site Setup	Confirm the seeder runs end-to-end for a second site without corrupting the first	Reliability Security
#90	P1	Onboarding & Site Setup	Make sure a second launch on the same session can't silently create a duplicate site	Reliability
#91	P1	Onboarding & Site Setup	Confirm the temp domain is assigned to the site record when the client picks a temporary URL	Client UX Reliability
#94	P1	Services	Prevent two services from sharing the same slug on the same site	Reliability
#108	P1	Products & Commerce	Make sure filtering products by category can't silently return wrong results	Reliability
#109	P1	Products & Commerce	Confirm disabled-module sites can't have their products scraped via the public endpoint	Security
#112	P1	Products & Commerce	Catch a future refactor that breaks the products list endpoint's visibility filter	Reliability Technical debt
#118	P1	Products & Commerce	Confirm featured products can't appear in the featured spotlight if they're still hidden	Client UX Reliability
#123	P1	Products & Commerce	Prevent two products from sharing the same slug on the same site	Reliability
#129	P1	Security & Seeding Safeguards	Confirm the seed gate can't be bypassed if SEED_ALLOWED is set to 'false' or empty	Security
#130	P1	Security & Seeding Safeguards	Make sure the add-on catalog seed can't run on production even if someone sets SEED_ALLOWED by mistake	Security
#131	P1	Security & Seeding Safeguards	Confirm visitors can't forge form submissions for a site they don't belong to	Security
#144	P1	User Management & Access Control	Confirm addSiteRole stays blocked even if superadmin status is set after roles already exist	Security

ID	PRIORITY	CATEGORY	IMPROVEMENT	TAGS
#145	P1	User Management & Access Control	Catch a superadmin partial update silently wiping roles on any site owner	Security Reliability
#146	P1	User Management & Access Control	Confirm role changes made through users.update are always recorded in the activity log	Security Reliability
#158	P1	Courses & Classes	Confirm a disabled-module site can't be unlocked by a direct courses.get call	Security
#184	P1	Permissions & Route Security	Confirm the module-access grid for site users also covers every role	Security
#185	P1	Permissions & Route Security	Confirm the module-access matrix stays complete when a dashboard module is added or removed	Security Infrastructure
#186	P1	Permissions & Route Security	Confirm the permissions preview always matches the role actually being saved	Client UX Security
#13	P2	User Management & Access Control	Confirm portal lockout and password flow can't be broken by a future change	Reliability Technical debt
#16	P2	Media & Images	Clean up legacy base64 images already in clients' media libraries	Technical debt Infrastructure
#30	P2	Infrastructure & DevOps	Make the client-website E2E suite actually run in CI against a deployed site	Infrastructure Reliability
#44	P2	Media & Images	Confirm the purge tool is safe to run when a site has thousands of records across multiple pages	Reliability
#58	P2	Infrastructure & DevOps	Make sure the workspace branch can't silently drift ahead of origin/main again	Infrastructure Technical debt
#84	P2	Performance & Build Quality	Confirm a dropped skeleton guard shows up as a failing test -- not a silent blank page	Technical debt Reliability
#93	P2	Onboarding & Site Setup	Seed placeholder services when onboarding creates a site with services enabled	Client UX
#114	P2	Products & Commerce	Confirm seeded product titles always carry the client's business name	Client UX
#192	P2	Onboarding & Site Setup	Add a CONTRIBUTING guide so new contributors always commit with the right identity	Infrastructure Technical debt

WAVE 3

Client Self-Service

Reduce FSTS manual support requirements

ID	PRIORITY	CATEGORY	IMPROVEMENT	TAGS
#45	P2	Media & Images	Let clients fix broken images directly from the nav badge -- not just see the count	Client UX Automation
#107	P2	Products & Commerce	Let visitors see which product categories exist before filtering	Client UX Revenue
#142	P2	Email & Notifications	Let admins re-send the welcome email to a pending user who hasn't signed in yet	Client UX Automation
#167	P2	Flyers & Marketing	Let clients duplicate an existing flyer instead of rebuilding from scratch	Client UX Automation
#168	P2	Flyers & Marketing	Confirm health notifications give a clear 'access denied' message to non-admin users	Client UX Security
#169	P2	Products & Commerce	Make sure client admins can manage their Square discounts without needing FSTS staff help	Client UX Revenue
#187	P2	User Management & Access Control	Let site owners invite dashboard users directly from their site -- without needing a super admin	Client UX Automation
#207	P2	Courses & Classes	Let admins fix a broken course link on an event from the warning banner -- not just see it	Client UX Automation

WAVE 4

UX & Performance

Loading states, navigation speed, accessibility, dashboard polish

ID	PRIORITY	CATEGORY	IMPROVEMENT	TAGS
#83	P2	Performance & Build Quality	Extend loading-skeleton guards to the remaining dashboard pages not yet covered	Client UX
#54	P3	Performance & Build Quality	Fix the circular vendor chunk warning so every production build is clean	Technical debt Infrastructure
#55	P3	Performance & Build Quality	Make page transitions feel instant by preloading chunks on hover	Client UX
#80	P3	Media & Images	Verify the image editor is keyboard-navigable and screen-reader friendly	Client UX
#110	P3	Onboarding & Site Setup	Remember that the getting-started nudge was dismissed so it doesn't reappear on every visit	Client UX
#111	P3	Onboarding & Site Setup	Confirm the placeholder nudge banner shows and disappears at the right times	Client UX
#176	P3	Sidebar & Navigation	Add a Role Permissions link to the per-site sidebar so admins can reach it from inside a site	Client UX

WAVE 5

Growth & Enhancement

Marketing, commerce, advanced automation, and non-blocking improvements

ID	PRIORITY	CATEGORY	IMPROVEMENT	TAGS
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8 improvements

User Management & Access Control

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#11	P1	W2	Let admins manually unlock a locked portal user account	Admins can unblock users without developer intervention, reducing support delays.	Client UX Security
2	#12	P1	W2	Stop other user management pages from leaking the full site list	Prevents unauthorized data exposure by scoping site lists to the correct context.	Security Reliability
3	#13	P2	W2	Confirm portal lockout and password flow can't be broken by a future change	Guards against regressions that could lock users out permanently.	Reliability Technical debt
4	#144	P1	W2	Confirm addSiteRole stays blocked even if superadmin status is set after roles already exist	Closes a privilege escalation window where late-granted superadmin could bypass role guards.	Security
5	#145	P1	W2	Catch a superadmin partial update silently wiping roles on any site owner	Prevents accidental data loss that could lock site owners out of their own accounts.	Security Reliability
6	#146	P1	W2	Confirm role changes made through users.update are always recorded in the activity log	Full audit trail for every permission change so admins can trace who changed what and when.	Security Reliability
7	#181	P1	W1	Catch a stale legacy role grant the moment a permission is renamed or removed	Prevents ghost permissions that outlive their intended purpose from accumulating over time.	Security
8	#187	P2	W3	Let site owners invite dashboard users directly from their site -- without needing a super admin	Eliminates the bottleneck of requiring FSTS staff involvement for every user invitation.	Client UX Automation

9 improvements

Permissions & Route Security

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#85	P0	W1	Make the guard check smart enough to catch null checks that don't show an access-denied message	Ensures users always see a clear 'access denied' message instead of a confusing blank screen.	Security
2	#86	P0	W1	Extend the access-denied guard requirement to any new route category added outside the sites folder	Future-proofs the app so new routes automatically inherit the correct access-denied behavior.	Security Infrastructure
3	#159	P0	W1	Confirm write-path guards work for CRM, forms, and media modules too	Closes gaps where CRM or media write operations could bypass module-access checks.	Security
4	#164	P0	W1	Confirm a client-role user is automatically blocked from every design-locked route	Enforces the client permission boundary so clients can't reach admin-only design tools.	Security
5	#171	P0	W1	Confirm the Footer, Email, CRM, and Payments pages can't be unlocked by a client user	Hardens the client boundary on critical business and payment configuration pages.	Security
6	#172	P0	W1	Catch a client user bypassing the navigation update and delete guards too	Ensures mutation guards on nav items cover all roles, not just the read-path checks.	Security
7	#184	P1	W2	Confirm the module-access grid for site users also covers every role	Guarantees no role is accidentally excluded from the module-access matrix.	Security
8	#185	P1	W2	Confirm the module-access matrix stays complete when a dashboard module is added or removed	Prevents new modules from launching without any permission boundaries set.	Security Infrastructure
9	#186	P1	W2	Confirm the permissions preview always matches the role actually being saved	Admins see exactly what they're granting before they save -- no surprise permission gaps.	Client UX Security

8 improvements

Media & Images

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#14	P0	W1	Let clients actually use their uploaded images across their whole site	Unlocks the media library as a shared asset pool, so clients get full value from every upload.	Client UX Revenue
2	#15	P1	W1	Prevent broken images when a media asset is deleted that's still in use	Eliminates embarrassing broken-image placeholders on client-facing websites.	Client UX Reliability
3	#16	P2	W2	Clean up legacy base64 images already in clients' media libraries	Reduces database size and speeds up media library load times across all client sites.	Technical debt Infrastructure
4	#44	P2	W2	Confirm the purge tool is safe to run when a site has thousands of records across multiple pages	Prevents partial deletes that would leave orphaned records when running cleanup at scale.	Reliability
5	#45	P2	W3	Let clients fix broken images directly from the nav badge -- not just see the count	Turns the broken-image badge into an actionable fix flow instead of a passive warning.	Client UX Automation
6	#80	P3	W4	Verify the image editor is keyboard-navigable and screen-reader friendly	Makes the image editor accessible to users who rely on keyboards or assistive technology.	Client UX
7	#81	P1	W1	Auto-generate alt text when images are uploaded so clients never have empty alt text	Improves SEO and accessibility on every client website automatically, with zero extra effort from clients.	Client UX Automation
8	#82	P1	W1	Confirm image picker flows work end-to-end in a real browser across every module	Catches browser-specific picker bugs before clients encounter them on live sites.	Reliability

11 improvements

Email & Notifications

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#21	P0	W1	Make failed welcome email deliveries visible so admins know when a new registrant didn't get their email	Admins can proactively follow up instead of discovering the issue after a registrant complains.	Client UX Reliability
2	#22	P1	W1	Warn admins when the lead alert recipient address looks wrong or bounces	Catches misconfigured alert emails before real leads go unnoticed.	Reliability Client UX
3	#23	P1	W1	Let admins send a test lead alert to confirm their notification email is working	One-click confidence check so admins know leads are flowing to the right inbox.	Client UX Automation
4	#26	P1	W1	Fix broken tests in the email suite that are silently hiding regressions	Restores reliable CI feedback so email bugs are caught before they reach production.	Technical debt Reliability
5	#27	P1	W1	Confirm the notification email override reaches the right address end-to-end	Guarantees the override setting actually redirects emails -- not just stores the address.	Reliability
6	#65	P0	W1	Let admins configure their Resend API key from the Email Config page -- not the Convex dashboard	Removes a technical barrier that currently requires developer access to configure email.	Client UX Automation
7	#66	P0	W1	Confirm a real email reaches the right inbox once Resend is configured for the first client	End-to-end proof that the Resend integration actually delivers before clients go live.	Reliability Client UX
8	#141	P1	W1	Confirm the welcome email fires when a new user is created -- and only then	Prevents duplicate welcome emails from confusing new users during re-invites or re-seeding.	Reliability
9	#142	P2	W3	Let admins re-send the welcome email to a pending user who hasn't signed in yet	Admins can help users who lost their welcome email without creating a new account.	Client UX Automation
10	#143	P1	W1	Make sure welcome emails link to the correct dashboard URL after deployment	New users land on the right domain -- not a broken or dev preview link.	Reliability

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
11	#202	P0	W1	Alert admins the moment a paid order confirmation email permanently fails to deliver	Paid customers always know their order went through -- failed emails surface immediately.	Revenue Reliability

8 improvements

Onboarding & Site Setup

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#72	P1	W2	Add an onboarding wizard so admins can seed a new client without touching the CLI	Non-technical staff can onboard new clients end-to-end through the dashboard UI.	Automation Client UX
2	#73	P1	W2	Confirm the seeder runs end-to-end for a second site without corrupting the first	Verifies multi-site isolation so onboarding a second client never affects existing data.	Reliability Security
3	#90	P1	W2	Make sure a second launch on the same session can't silently create a duplicate site	Prevents orphaned duplicate site records from confusing admins and fragmenting data.	Reliability
4	#91	P1	W2	Confirm the temp domain is assigned to the site record when the client picks a temporary URL	New clients can immediately preview their site at the expected URL after onboarding.	Client UX Reliability
5	#93	P2	W2	Seed placeholder services when onboarding creates a site with services enabled	New sites launch with example content rather than an empty, confusing services section.	Client UX
6	#110	P3	W4	Remember that the getting-started nudge was dismissed so it doesn't reappear on every visit	Eliminates a repetitive UI annoyance that makes the dashboard feel unpolished.	Client UX
7	#111	P3	W4	Confirm the placeholder nudge banner shows and disappears at the right times	Makes the onboarding experience feel intentional -- guides new users, then gets out of the way.	Client UX
8	#192	P2	W2	Add a CONTRIBUTING guide so new contributors always commit with the right identity	Prevents messy commit history and identity-check failures from new team members.	Infrastructure Technical debt

8 improvements

Products & Commerce

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#107	P2	W3	Let visitors see which product categories exist before filtering	Visitors can browse categories without guessing what's available, boosting discoverability.	Client UX Revenue
2	#108	P1	W2	Make sure filtering products by category can't silently return wrong results	Prevents clients from unknowingly showing incorrect product sets to customers.	Reliability
3	#109	P1	W2	Confirm disabled-module sites can't have their products scraped via the public endpoint	Protects product data from sites that haven't purchased the commerce module.	Security
4	#112	P1	W2	Catch a future refactor that breaks the products list endpoint's visibility filter	Automated check that draft/hidden products can never leak to the public storefront.	Reliability Technical debt
5	#114	P2	W2	Confirm seeded product titles always carry the client's business name	Demo products feel branded from day one, giving clients a realistic preview of their store.	Client UX
6	#118	P1	W2	Confirm featured products can't appear in the featured spotlight if they're still hidden	Prevents draft products from accidentally appearing in premium homepage real estate.	Client UX Reliability
7	#123	P1	W2	Prevent two products from sharing the same slug on the same site	Eliminates broken product URLs caused by slug collisions that route to the wrong item.	Reliability
8	#169	P2	W3	Make sure client admins can manage their Square discounts without needing FSTS staff help	Clients can run promotions independently, removing a support dependency on FSTS staff.	Client UX Revenue

3 improvements

Payments & Checkout

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#203	P0	W1	Confirm Square payments still process correctly end-to-end after the idempotency changes	Validates that the idempotency hardening didn't break the actual checkout flow for real customers.	Revenue Reliability
2	#204	P0	W1	Harden the provider-agnostic payment webhook with the same idempotency and email-state guards	Ensures future payment integrations can't double-charge customers or skip order confirmation emails.	Revenue Reliability Security
3	#206	P0	W1	Block checkout when a course or event price is missing or corrupt so no broken order reaches Square	Prevents \$0 or malformed orders from reaching Square, protecting clients from disputes and chargebacks.	Revenue Reliability

7 improvements

Courses & Classes

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#157	P1	W1	Confirm courses stay hidden when the agency switches the module off platform-wide	Module toggles work at the platform level -- clients can't circumvent a module being disabled.	Security Reliability
2	#158	P1	W2	Confirm a disabled-module site can't be unlocked by a direct courses.get call	API-level guard prevents clever URL manipulation from bypassing module restrictions.	Security
3	#189	P1	W1	Confirm lifecycle alerts in the edit dialog always reflect live registration data	Admins editing a class see accurate enrollment counts, not stale cached data.	Client UX Reliability
4	#193	P0	W1	Make sure a client with a stale session can't register for a class that's already been cancelled	Prevents ghost registrations where customers pay for a class that no longer exists.	Revenue Reliability
5	#194	P1	W1	Catch a flyer that stays published after its linked class is deleted outright	Orphaned flyers advertising deleted classes are automatically unpublished to avoid customer confusion.	Client UX Reliability
6	#195	P1	W1	Confirm the lifecycle clock is registered so tick can never be silently skipped	Class status transitions (open -> full -> closed) happen on schedule rather than getting stuck.	Reliability Automation
7	#207	P2	W3	Let admins fix a broken course link on an event from the warning banner -- not just see it	Turns a passive warning about broken course links into a one-click repair action.	Client UX Automation

3 improvements

Services

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#92	P0	W1	Confirm services created in the dashboard actually appear on the public website	Closes the gap where a service looks correct in the dashboard but never surfaces on the live site.	Client UX Reliability
2	#94	P1	W2	Prevent two services from sharing the same slug on the same site	Avoids URL collisions that cause the wrong service page to load for customers.	Reliability
3	#137	P1	W1	Seed services data for Corsair so the live services section appears on the website	Corsair's website displays a real services section immediately after seeding, no manual entry needed.	Client UX

2 improvements

Flyers & Marketing

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#167	P2	W3	Let clients duplicate an existing flyer instead of rebuilding from scratch	Dramatically speeds up flyer creation for recurring events -- clients copy and adjust instead of starting over.	Client UX Automation
2	#168	P2	W3	Confirm health notifications give a clear 'access denied' message to non-admin users	Non-admin users get a clear boundary message instead of an unexplained error.	Client UX Security

1 improvements

Sidebar & Navigation

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#176	P3	W4	Add a Role Permissions link to the per-site sidebar so admins can reach it from inside a site	Eliminates navigation friction -- admins can jump directly to role settings from any site context.	Client UX

4 improvements

Performance & Build Quality

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#54	P3	W4	Fix the circular vendor chunk warning so every production build is clean	Clean builds with no warnings make it easier to spot real problems in CI output.	Technical debt Infrastructure
2	#55	P3	W4	Make page transitions feel instant by preloading chunks on hover	Dashboard navigation feels snappier -- chunks are ready before the user clicks.	Client UX
3	#83	P2	W4	Extend loading-skeleton guards to the remaining dashboard pages not yet covered	Eliminates blank-flash loading states on pages that currently show nothing while data loads.	Client UX
4	#84	P2	W2	Confirm a dropped skeleton guard shows up as a failing test -- not a silent blank page	Prevents future regressions where a missing skeleton silently degrades the loading experience.	Technical debt Reliability

2 improvements

CMS & Live Data

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#135	P0	W1	Confirm a price change in the dashboard actually updates the Corsair website within 60 seconds	Clients can trust the dashboard price is what customers see -- no manual cache purge needed.	Client UX Reliability
2	#136	P1	W1	Extend live CMS data to the course detail pages so per-course pricing is also accurate	Detail pages reflect real pricing, preventing customers from seeing stale rates.	Client UX Reliability

3 improvements

Security & Seeding Safeguards

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#129	P1	W2	Confirm the seed gate can't be bypassed if SEED_ALLOWED is set to 'false' or empty	Prevents accidental data seeding on production environments from a misconfigured env var.	Security
2	#130	P1	W2	Make sure the add-on catalog seed can't run on production even if someone sets SEED_ALLOWED by mistake	Belt-and-suspenders protection against overwriting live add-on catalog data.	Security
3	#131	P1	W2	Confirm visitors can't forge form submissions for a site they don't belong to	Closes a cross-site form injection vector where a malicious visitor could pollute another site's CRM.	Security

6 improvements

Infrastructure & DevOps

#	ID	PRIORITY	WAVE	WHAT IT DOES	WHY IT MAKES THE DASHBOARD BETTER	TAGS
1	#20	P1	W1	Make sure Vercel deploys cleanly against the rewritten commit history	Unblocks deployments that were failing because of the history rewrite required for identity cleanup.	Infrastructure
2	#30	P2	W2	Make the client-website E2E suite actually run in CI against a deployed site	Catches client-facing website regressions in CI instead of after a client reports them.	Infrastructure Reliability
3	#58	P2	W2	Make sure the workspace branch can't silently drift ahead of origin/main again	Prevents the confusing scenario where local and remote history diverge without warning.	Infrastructure Technical debt
4	#61	P1	W1	Confirm www.fstsclientsystem.com actually redirects once the Vercel alias is added	Clients can reach the dashboard at the canonical www address -- not just the bare domain.	Infrastructure Client UX
5	#62	P1	W2	Turn on the branch protection rule so the identity check can't be bypassed on GitHub	Enforces commit identity standards at the repository level -- no accidental bypass via direct push.	Infrastructure Security
6	#211	P0	W1	Confirm clients can actually sign in once the Cloudflare DNS fix is applied	Verifies the DNS fix resolves the sign-in issue for clients on all networks.	Client UX Infrastructure

ARCHITECTURAL FOUNDATION

FSTS-WOS(TM) Multi-Site & E-Commerce Architecture Requirements

?? Core Rule

Do not design the FSTS-WOS(TM) around Corsair specifically. Every feature, model, permission, automation, CMS module, API, and component must be evaluated for reuse across many independent client websites.

? Client Design Lock

Clients manage approved content, media, products, services, classes, forms, and SEO. They cannot modify source code, structural layouts, design system, backend, or deployment infrastructure. FSTS controls all templates.

? Multi-Tenant Isolation

Every record must belong to the appropriate tenant. A client must never read, modify, search, export, or indirectly discover another client's products, orders, customers, payments, registrations, media, leads, or configuration.

? Provider-Agnostic Commerce

Square is the first supported payment provider, but the architecture must allow additional providers later. Use provider abstraction for payments, refunds, customers, orders, discounts, and webhooks. Store provider IDs separately from FSTS IDs.

? Success Test

Can we onboard a completely different business -- including a full e-commerce business -- without changing the core dashboard code? The answer must eventually be YES. Corsair proves #1. Website #2 proves repeatability. An e-commerce client proves platform flexibility. #10 proves scalability.

? Checklist for Every Task

1. Is this reusable across tenants?
2. Does it assume Corsair-specific behavior?
3. Does it work for non-Corsair businesses?
4. Does it work when the relevant module is disabled?
5. Does it maintain tenant isolation?
6. Does it respect the client design lock?
7. Could it work for an e-commerce site?
8. Does it scale to dozens or hundreds of sites?

Supported Business Types (Platform Must Handle All)

Service Businesses

E-Commerce Stores

Training Companies

Course/Class Businesses

Event Businesses

Membership Organizations

Professional Services

Restaurants & Hospitality

Nonprofits

Contractors

Retail Businesses

Product + Service Hybrids

Digital Products

Physical Products

Appointments/Bookings

Subscriptions/Memberships

ROADMAP SUMMARY

Execution Statistics & Top Priorities

83

Total Improvements

18

P0 Critical

41

P1 Required

18

P2 Important

6

P3 Enhancement

36

Wave 1 Items

32

Wave 2 Items

15

Categories

Top 10 Highest-Value Immediate Actions

#	ID	IMPROVEMENT	PRIORITY	RATIONALE
1	#211	Confirm clients can actually sign in once the Cloudflare DNS fix is applied	P0	Clients cannot sign in -- blocking all revenue
2	#203	Confirm Square payments still process correctly end-to-end after the idempotency changes	P0	Validates payment flow after idempotency changes
3	#206	Block checkout when a course or event price is missing or corrupt so no broken order reaches Square	P0	Stops \$0/corrupt orders reaching Square
4	#204	Harden the provider-agnostic payment webhook with the same idempotency and email-state guards	P0	Protects all future payment integrations
5	#92	Confirm services created in the dashboard actually appear on the public website	P0	Services not appearing on live website (#1 CMS gap)
6	#202	Alert admins the moment a paid order confirmation email permanently fails to deliver	P0	Paid customers must receive order confirmation
7	#65	Let admins configure their Resend API key from the Email Config page -- not the Convex dashboard	P0	Email config requires developer access today
8	#193	Make sure a client with a stale session can't register for a class that's already been cancelled	P0	Ghost registrations = payments for cancelled classes
9	#85	Make the guard check smart enough to catch null checks that don't show an access-denied message	P0	Blank screens instead of access-denied messages
10	#135	Confirm a price change in the dashboard actually updates the Corsair website within 60 seconds	P0	Price changes must reflect within 60 seconds